
A research agent for aging study identifies previously unknown aging mechanisms

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Abstract

Elucidating the causal drivers of aging is impeded by its inherent stochasticity, multi-scale complexity, and profound cellular heterogeneity, compounded by an unmanageable volume of biomedical data. To overcome these barriers, we present PersonaAI, an interactive agentic-AI framework designed to function as a “digital co-scientist.” By integrating literature-based reasoning with autonomous *in silico* validation, PersonaAI synthesizes over 560,000 aging-related publications via retrieval-augmented generation (RAG) to translate mechanistic hypotheses directly into executable single-cell analytical workflows using natural language. We demonstrated the system’s capacity for genuine inference beyond simple information retrieval through a rigorous temporal cutoff strategy: hypotheses generated solely using pre-2020 literature were subsequently substantiated by post-2021 discoveries. In application, the system uncovered a novel male-specific decline in adipose stem and progenitor cells driven by vascular niche deterioration and disrupted VEGF–VEGFR signaling. This mechanism is autonomously validated further by PersonaAI’s single-cell RNA-seq agent. These results establish PersonaAI as a scalable, generalizable platform that accelerates discovery in aging biology by augmenting human intuition with autonomous, data-driven validation.

1. Introduction

Aging is a universal, multi-scale process driving the trajectory of life from development to decline (López-Otín et al., 2023). Unlike acute disease states, aging is defined by profound stochasticity and temporal latency. It manifests as an asynchronous accumulation of damage where neighboring cells in identical tissues may exhibit vastly different molecular signatures (Enge et al., 2017; Martinez-Jimenez et al., 2017). Because this process is driven by stochastic entropy rather than deterministic programming, it represents one of the most intractable analytical challenges in modern biology. This complexity is compounded by the vast temporal scale of mammalian aging, which imposes severe logistical barriers. Establishing rigorous causality between early-life molecular events and late-life phenotypes requires tracking trajectories over decades.

To manage this complexity, the field has prepared large-scale datasets using high-throughput technologies. Comprehensive single-cell transcriptomic atlases (Almanzar et al., 2020; Zhang et al., 2025) now provide a foundational landscape of aging signatures across murine and human lifespans, enabling the *in silico* interrogation of temporal heterogeneity. However, these high-dimensional repositories have created a secondary barrier of interpretation. The ability to distinguish biologically meaningful causal patterns from technical artifacts or “passenger” noise now requires a depth of multi-disciplinary expertise that is impossible to scale through human effort alone (Wilkinson et al., 2016).

The engine of scientific discovery is fueled by the recursive synthesis of accumulated knowledge and empirical validation. Traditionally, researchers have relied on their knowledge to generate hypotheses and reason through potential biological mechanisms. However, with over 560,000 articles indexed in PubMed as of 2025 in aging, the information landscape is too vast for any individual researcher to navigate (Bormmann & Mutz, 2014). This challenge is further intensified by the inherent interdisciplinary nature of gerontology, which demands a simultaneous mastery of evolutionary biology, immunology, metabolism, and neuroscience (Campisi et al., 2019).

To overcome the cognitive and bioinformatic bottleneck, we propose PersonaAI. PersonaAI is an agentic-AI framework designed to function as a digital co-pilot (Boiko et al., 2023; Wang et al., 2023) for aging research. Rather than acting as a passive database, PersonaAI engages in an interactive reasoning loop with the researcher to synthesize existing literature into novel, testable hypotheses. Once a biological mechanism is proposed, the system's orchestrator autonomously constructs and executes dynamic analytical pipelines, rigorously validating the hypothesis in silico by querying high-dimensional single-cell aging atlases.

To provide the necessary biological context, PersonaAI is anchored by a comprehensive knowledge base of over 560,000 aging-related publications. Rather than functioning as a standard search engine, the system utilizes a retrieval-augmented generation (RAG) architecture to reconstruct and synthesize information from diverse biomedical resources (Pan et al., 2024). This allows the system to transcend simple keyword retrieval and performs semantic reasoning to bridge between disciplines, identifying indirect causal links. Operating as a true digital co-scientist, PersonaAI engages in an iterative dialectic with the researcher. Through this interactive feedback loop, the agent assists the human partner in refining broad inquiries into concrete, mechanistically sound, and data-testable hypotheses, effectively accelerating the path from literature review to experimental design. Once a hypothesis is crystallized, PersonaAI deploys a specialized fleet of multi-agent subroutines to orchestrate the in-silico validation process. This modular architecture allows the system to autonomously requisition high-dimensional data from the Mouse Aging Atlas and execute complex analytical pipelines.

PersonaAI represents a significant evolution beyond pioneering autonomous frameworks such as Virtual Lab and AI Co-scientist (Boiko et al., 2023; Gottweis et al., 2025; Swanson et al., 2025). While these autonomous partners offer speed and scalability, they face a fundamental efficiency bottleneck: they often rely on unconstrained stochastic exploration, generating vast numbers of hypotheses through a computationally expensive guess-and-check approach. Platforms such as AI Co-scientist (M. Bran et al., 2024) do not possess an integral mechanism for empirical validation. In contrast, PersonaAI diverges from the "brute force" paradigm by operating as a strategic co-pilot. By maintaining human intuition within the reasoning loop, the system effectively constrains the search space to biologically plausible mechanisms. Technically, while earlier systems focused on domains with established predictive models, such as protein folding (Jumper et al., 2021; Swanson et al., 2025), PersonaAI incorporated model context protocols (MCPs) to translate unstructured natural language into executable bioinformatic workflows, enabling researchers to perform complex in silico validation of aging mechanisms without specialized coding expertise.

The predictive validity of this architecture is substantiated by our retrospective analysis using temporal cut-off, which serves as a blinded benchmark for prospective discovery. PersonaAI newly identified that the male-specific regenerative decline of Adipose Stem and Progenitor Cells (ASPCs) is not merely intrinsic but is actively orchestrated by the deterioration of the vascular endothelial niche.

PersonaAI functions as an active collaborative partner that engages the researcher in a structured reasoning loop. The system guides logical refinement from broad biological inquiries into precise, mechanistically grounded hypotheses and further assessment based on genomic data.

2. Method Overview

To implement the proposed digital co-scientist, PersonaAI, we integrated a multi-agent architecture with a specialized Knowledge Graph constructed from over 560,000 aging-related publications. The system operates through two primary components: (1) an Orchestrator Agent, which utilizes a Retrieval-Augmented Generation (RAG) module to synthesize mechanistic hypotheses from the structured knowledge base; and (2) a Code Agent, which autonomously generates and executes Python scripts (e.g., using Scanpy) to validate these hypotheses against high-dimensional single-cell datasets.

Detailed descriptions of the agentic workflow, KG construction, and data processing pipeline are provided in Appendix A. Supplementary figures and tables supporting our in silico validation are available in Appendix B.

3. Result

Construction of an Interactive Agentic Framework for Aging Hypothesis Generation and Validation

Aging research faces critical challenges driven by biological stochasticity and temporal latency, which obscure causal mechanisms due to cellular heterogeneity, and the data overload from synthesizing from 565,634 publications (PMC: 212,041 and PubMed: 353,593). To address these issues, we developed PersonaAI, a dual-phase agentic framework (Figure 1).

Unlike standard Large Language Models (LLMs) that primarily retrieve information, PersonaAI acts as an active research partner through a Human-Agent Interaction Loop (Figure 1A). Rather than acting as a passive knowledge store, the system employs an agentic architecture designed to emulate the iterative nature of scientific reasoning. Through continuous, iterative dialogue, the researcher provides high-level biological guidance, which the system uses to navigate the vast search space of aging literature. This process facilitates knowledge synthesis by bridging disparate datasets and publications to identify hidden biological connections. By actively engaging the user, the system functions to distill fragmented information, effectively narrowing down broad, abstract concepts into precise, mechanistically grounded, and testable hypotheses. By keeping the human expert in the loop, PersonaAI ensures that generative creativity remains constrained by biological plausibility and clinical relevance (Figure 1A).

To move from theory to validation, the framework employs a “Hypothesis Evaluation Agent” (Figure 1B). Instead of following a fixed script, this agent dynamically coordinates specialized tools to test each hypothesis. For instance, the Single-cell Atlas MCP queries a comprehensive atlas of 25 million cells pre-processed with rigorous quality control, batch correction, and cell-type annotations, ensuring that biological noise is managed through high-quality, large-scale data integration. The agent then directs sub-tools for clustering and pathway analysis, seamlessly converting abstract hypotheses into quantitative results (See Methods).

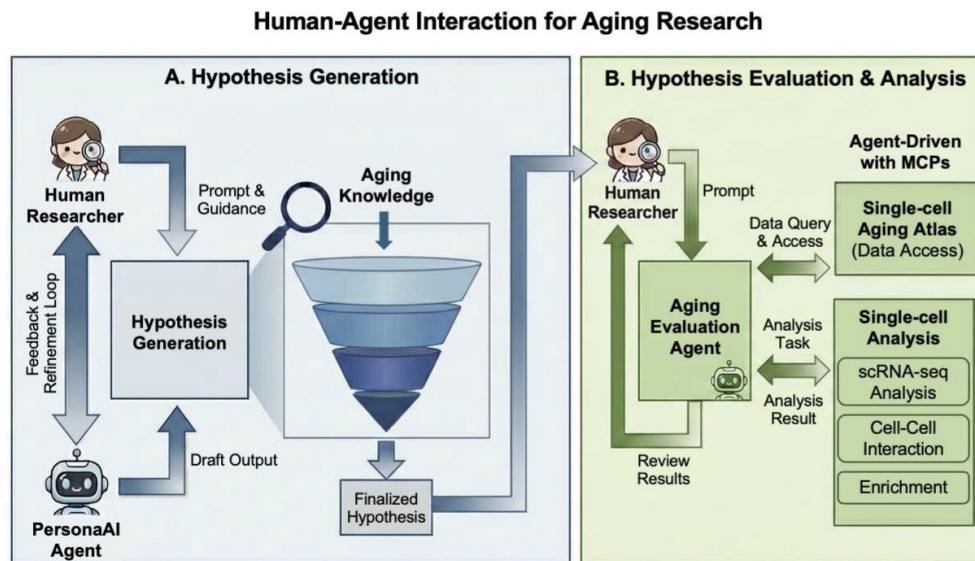


Figure 1: Framework of PersonaAI for autonomous hypothesis generation and in-silico validation in aging research.

(A) Schematic of the hypothesis generation phase featuring a Human-Agent interaction loop for literature synthesis and recursive refinement. (B) Evaluation and analysis phase driven by Agent-driven MCPs for autonomous data access and execution of specialized bioinformatic tasks.

Validation of Hypothesis Generation Capability via Temporal Cutoff Strategy

To rigorously evaluate PersonaAI’s capacity to generate scientifically valid hypotheses rather than merely retrieving existing facts, we implemented a retrospective validation strategy using a temporal

cutoff. We restricted the agent’s external knowledge to literature published prior to 2020 and tasked it with generating hypotheses for established aging hallmarks. To ensure that generated hypotheses represent novel inferences rather than reproduction of existing knowledge, each hypothesis was screened against the same pre-2020 corpus to verify the absence of prior reporting. Only hypotheses without direct precedent were retained for subsequent validation. The filtered outputs were then cross-referenced against the corpus of literature published between 2021 and 2025 to verify whether the agent autonomously constructed logic that aligns with future discoveries.

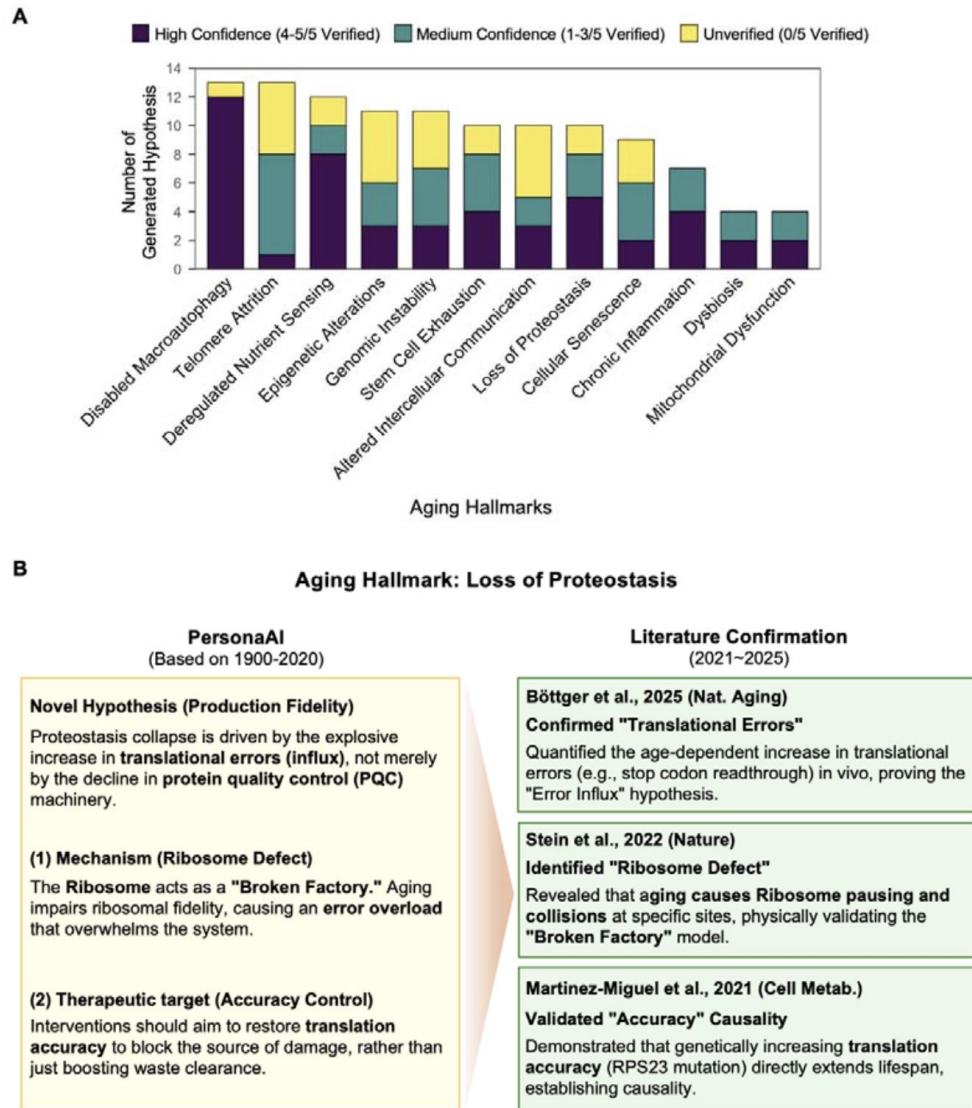


Figure 2. Benchmarking of PersonaAI discovery capability using a temporal cutoff strategy.

(A) Distribution of 114 generated hypotheses across 12 aging hallmarks, categorized by ensemble verification confidence against 2021–2025 literature. (B) Case study of “Loss of Proteostasis” hallmark showing predicted ribosomal translational infidelity mechanisms and subsequent manual validation using post-2020 discoveries

We produced 114 hypotheses across the spectrum of 12 aging hallmarks. To quantify the robustness of these predictions, the hypotheses were evaluated 5 times (Supplementary Figure 1) and categorized into three levels of reliability: ‘High Confidence’ (verified in 4 to 5 out of 5 independent runs), ‘Medium Confidence’ (verified in 1 to 3 runs), and ‘Unverified’ (0 runs). Intriguingly, we observed both verifiable and unverifiable hypotheses from the recent studies in most of the hallmarks. This suggests that the system’s logic-driven outputs are not random hallucinations but are grounded in reproducible biological reasoning.

It is also of note that there are unverified hypotheses in this screening. One illustrative case is observed in the domain of “Loss of Proteostasis” (Figure 2B). While the pre-2020 consensus largely attributed proteostasis collapse to a decline in degradation machinery (e.g., autophagy or proteasome failure), PersonaAI synthesized a hypothesis centered on production fidelity. It further delineated a specific mechanism related with the ribosome defect. It further suggested that restoring translation accuracy as the therapeutic targets (Figure 2B). We conducted a manual verification to ensure that these specific mechanisms were entirely absent from the scientific literature published prior to 2020. Interestingly, the suggested hypothesis, mechanisms, and the therapeutic targets were observed in recent study. Böttger et al. quantified the age-dependent increase in translational errors (e.g., stop codon readthrough) in vivo in 2025 (Böttger et al., 2025). Stein et al. revealed that aging induces specific ribosome pausing and collisions, providing mechanistic support consistent with the predicted impairment in 2022 (Stein et al., 2022). Furthermore, Martinez-Miguel et al. showed that genetically increasing translation accuracy (via an RPS23 mutation) can extend lifespan in 2021 (Stein et al., 2022) (Figure 2B).

PersonaAI also synthesized a hypothesis that senescent cells secrete bioactive lipids that act as primary drivers of inflammation, expanding the definition of Senescence-Associated Secretory Phenotype (SASP) (Supplementary Figure 2). We also performed manual verification to ensure that it is not studied before 2020 but shown in recent studies (Byrns et al., 2024; Narzt et al., 2021; Wiley et al., 2021).

Collectively, these findings demonstrate that PersonaAI can bridge fragmented pre-2020 evidence and post-2020 findings in a way that yields testable mechanistic hypotheses. The results confirm that the framework possesses the inferential capacity to transcend simple information retrieval, establishing its utility as a powerful engine for de novo discovery in aging research.

Uncovering Cellular Drivers of Aging Phenotypes: The Identification of Senescent Cirbp+ Hepatocytes.

To evaluate the system’s capacity for cross-scale synthesis, we challenged PersonaAI to resolve the cellular context of genes previously identified only in lower-resolution studies. Specifically, we tasked the agent to identify aging-associated genes reported in bulk tissue repositories that currently lack single-cell resolution and to propose a plausible cellular mechanism for their action (Figure 3A). This discovery workflow was executed using the structured prompt logic detailed in Supplementary Table 3.

PersonaAI picked Cold Inducible RNA Binding Protein (CIRBP) as a target for further study using scRNAseq. To obtain it, PersonaAI linked the “Phenotypic Evidence” of systemic CIRBP upregulation observed in human skeletal muscle bulk transcriptomics (Su et al., 2015) with the “Mechanistic Evidence” of liver-specific stress signals reported in independent studies (Izgi et al., 2022). CIRBP is a stress-responsive RNA-binding protein that has been implicated in inflammatory signaling and reported to show age-associated regulation in multiple contexts (Barth et al., 2021; Qiang et al., 2013). Based on these evidences, PersonaAI formulated two competing hypotheses to distinguish the most plausible cellular driver mechanism. 1) Hypothesis A (Hepatocyte Stress): CIRBP induction represents an intrinsic defensive response of hepatocytes against oxidative or proteotoxic stress and 2) Hypothesis B (Immune Activation): the signal stems from activated resident macrophages, specifically Kupffer cells, reflecting chronic inflammaging. To further clarify the hypothesis, PersonaAI extracted the longitudinal expression profile of Cirbp to assess lineage-specific trends. This initial screening revealed that while myeloid populations displayed minimal changes, the hepatocyte lineage exhibited a robust, age-dependent upregulation of Cirbp (Wilcoxon test, $p < 0.00001$) (Figure 3B). Because the atlas analysis indicated that age-associated Cirbp upregulation in hepatocytes, we focused subsequent analyses on hepatocyte-intrinsic stress programs rather than immune-driven explanations: “Do these Cirbp-high hepatocytes exhibit specific senescence characteristics?” (Figure 3C). In response, PersonaAI downsampled the hepatocyte population, performed unsupervised clustering, and identified a distinct subpopulation related with highest levels of Cirbp expression (C4 in Figure 3D, 3E). Temporal distribution analysis demonstrated that cells in C4 are rarely found in young mice (3 months) but undergo a dramatic expansion during aging, becoming a dominant subpopulation by 23 months (Figure 3E, F). Finally, the agent performed deep phenotyping to characterize the molecular identity of this Cirbp-enriched cluster. The analysis revealed that C4 is defined by a specific intrinsic stress signature rather than general inflammation, exhibiting upregulation of canonical senescence markers (e.g., Cdkn1a,

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Trp53, Cdkn2c) and key Endoplasmic Reticulum (ER) stress regulators (e.g., Hspa5, Ern1, Atf4) (Figure 3G). Collectively, these results demonstrate that PersonaAI serves as a powerful discovery engine that augments the researcher by synthesizing novel findings through an integrated cycle of hypothesis generation and rigorous genomic validation. By integrating literature-based reasoning with high-dimensional single-cell analysis, the framework successfully transforms fragmented biological data into actionable, evidence-based insights.

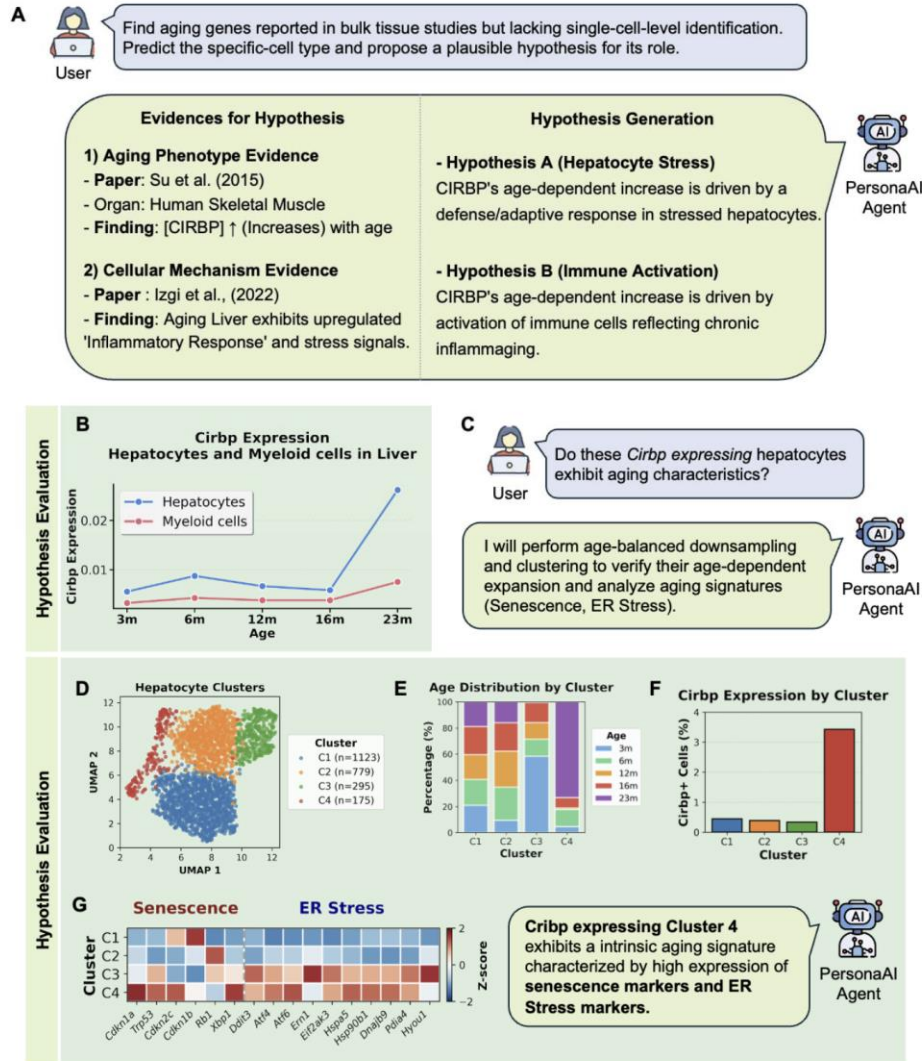


Figure 3. Identification of senescent Cirbp+ hepatocytes through cross-scale synthesis.

(A) Discovery workflow bridging bulk tissue findings with single-cell level mechanistic identification. (B) Expression profile of Cirbp in liver hepatocyte and myeloid lineages, indicating hepatocyte-specific age-dependent upregulation. (C) Narrative of AI-human interactive inquiry regarding the aging characteristics of Cirbp-high hepatocytes. (D) UMAP visualization of hepatocyte clusters (C1–C4) derived from unsupervised clustering. (E) Cluster proportion changes across age groups (3m–23m) highlighting the selective expansion of cluster C4. (F) Relative Cirbp expression across clusters suggesting cluster C4 as a potential driver of age-associated induction. (G) Molecular signatures of senescence (e.g., Cdkn1a, Trp53) and ER stress (e.g., Hspa5, Atf4) characterizing the Cirbp-enriched cluster C4.

PersonaAI identifies the vascular niche as a driver of male-specific ASPC decline

To facilitate the strategic distillation of research inquiries, we implemented a hierarchical Meta-Prompting architecture that operates through three-step hierarchical phases to concretize the hypothesis (Method). Drawing upon the principles of Step-Back Prompting (Zheng et al., 2024), this

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framework compels the agent to first abstract the initial query into fundamental biological principles before converging on specific molecular targets. This top-down approach ensures that the resulting hypotheses are grounded in broad physiological contexts while maintaining the precision required for experimental validation.

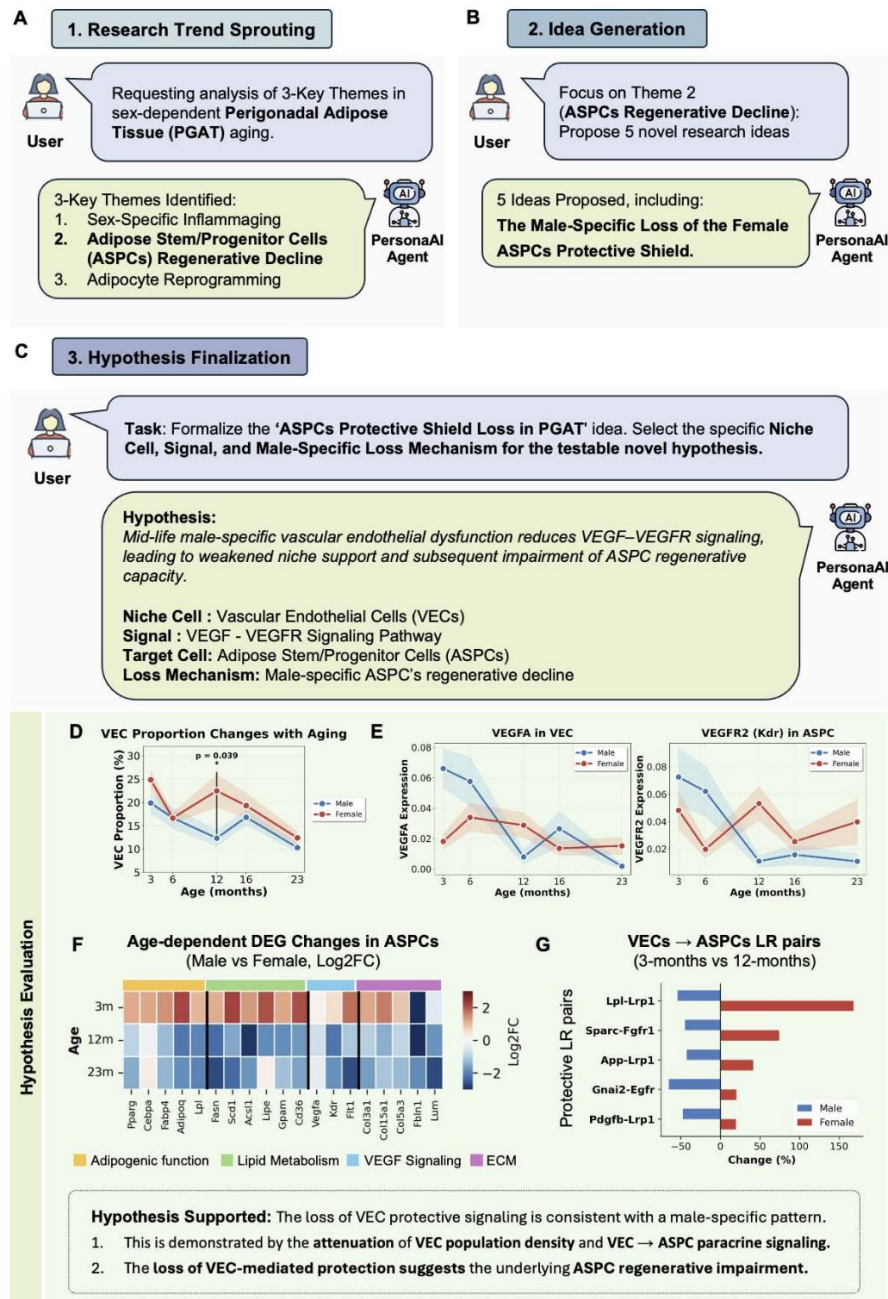


Figure 4. Uncovering the vascular niche as a driver of male-specific ASPC decline.

(A) Research trend sprouting phase identifying three high-impact themes in sex-dependent PGAT aging. (B) Idea generation phase proposing mechanistic narratives for male-specific regenerative impairment. (C) Hypothesis finalization model integrating the VEGF-VEGFR signaling axis and niche dysfunction. (D) Age-dependent reduction in VEC population proportion specifically in the male cohort. (E) Expression levels of ligand Vegfa in VECs comparing male and female trajectories. (F) Receptor expression of Vegfr2 (Kdr) in recipient ASPCs, showing parallel decline in males. (G) Ligand-receptor (LR) interaction analysis between VECs and ASPCs indicating a male-specific loss of pro-regenerative paracrine signaling.

This prompt (Supplementary Table 4) led us to 3 themes in sex-dependent perigonadal adipose tissue (PGAT) aging. We specifically targeted this tissue because, despite the profound sexual dimorphism well-documented in metabolic aging (Mauvais-Jarvis et al., 2020), the specific cellular determinants and niche-derived signals driving these divergent trajectories remain largely elusive even in recent transcriptomic landscapes (Emont et al., 2022).

The investigation was initiated by a broad query to “analyze key themes in sex-dependent PGAT aging.” In response, PersonaAI proposed three distinct themes. From these options, we selected “Theme 2: Adipose Stem/Progenitor Cells (ASPCs) Regenerative Decline” for further characterization, prioritizing it over the well-saturated topic of general inflammaging to maximize novelty (Figure 4A, Supplementary Table 5).

Subsequently, focusing on the selected theme, the agent generated five candidate research ideas. We evaluated these proposals based on biological plausibility and alignment with aging hallmarks and selected ‘Idea3: The Male-Specific Loss of the Female ASPCs Protective Shield’ as the most compelling candidate due to its strong potential connection to the “Altered Intercellular Communication” of aging hallmark (Figure 4B, Supplementary Table 6). To translate this abstract narrative into a testable mechanism, we instructed the PersonaAI to formalize the selected idea. PersonaAI structured the final hypothesis by explicitly defining the biological entities involved. The system postulated that the regenerative decline in male ASPCs is not intrinsic but is actively driven by the deterioration of the Vascular Endothelial Cell (VEC) niche and the consequent loss of the VEGF-VEGFR signaling axis (Figure 4C, Supplementary Table 7).

To validate the AI-generated hypothesis positing a male-specific collapse of the vascular niche, we executed a targeted validation pipeline on the sex-stratified PGAT single-cell atlas using PersonaAI. We observed that the proportion of Vascular Endothelial Cells (VECs) remained stable in aging females. However, aging males exhibited a significant and progressive reduction in VEC abundance ($p=0.039$) (Figure 4D). This structural decline suggests a physical loss of the supportive niche specifically in the old male mice.

Next, we interrogated the molecular integrity of the hypothesized signaling axis. Consistent with the population loss, VECs in aging males displayed a marked downregulation of the ligand *Vegfa*, whereas female VECs maintained robust expression levels throughout the lifespan (Figure 4E). Concurrently, the recipient ASPCs in males showed a parallel decline in the expression of the primary receptor, *Kdr* (VEGFR2), suggesting that VEGF signal transmission is likely reduced due to the concurrent loss of both the ligand and its receptor (Figure 4F).

Finally, to quantify the functional consequences of these alterations, the agent performed a ligand–receptor (L–R) interaction analysis between VECs and ASPCs. This analysis indicated a marked weakening of protective paracrine signaling in aging males, with several pro-regenerative and metabolic interactions—including *Lpl–Lrp1*, *Sparc–Fgfr1*, and *Pdgfrb–Lrp1*—showing reduced interaction strength in aging males compared to young controls, a pattern that was notably absent in the female cohort (Figure 4G). Collectively, these computational results are consistent with the PersonaAI-derived hypothesis that male ASPC decline may be driven, at least in part, by deterioration of vascular niche support and reduced VEGF-mediated crosstalk.

To ensure that the observed sex-dimorphic patterns were not driven by technical artifacts, we performed a manual validation of the analytical pipeline. We verified that the scRNA-seq integration achieved effective batch correction while preserving biological heterogeneity across all age and sex groups (Supplementary Figure 3A–C). Furthermore, examination of the differential expression profiles confirmed that the identified signatures were statistically robust and distinct from technical noise (Supplementary Figure 4A and B). Taken together, these validation steps support the reliability of the AI-derived mechanistic model.

4. Discussions

The emergence of Agentic AI marks a fundamental paradigm shift, transitioning LLMs from passive conversational interfaces to autonomous scientific engines capable of independent reasoning, strategic decision-making, and active discovery (Gibney, 2025). Unlike traditional chatbots whose function is limited to reactive information retrieval, agentic systems operate with autonomy, actively participating the scientific process.

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We engineered PersonaAI to function as an interactive cognitive co-pilot specifically optimized for the complexities of aging research. Far exceeding the capabilities of a standard knowledge retrieval tool, the system is designed to synthesize novel hypotheses by bridging disparate conceptual domains across the aging literature. As demonstrated in our validation studies (Figures 3 and 4), PersonaAI acts as a rigorous scientific partner. It engages the researcher in a dialectic process to refine theoretical models and immediately validate them against high-dimensional scRNAseq datasets.

Distinct from fully autonomous hypothesis generators (Boiko et al., 2023; Lu et al., 2024), PersonaAI is engineered with a human-centric design philosophy. In this framework, the researcher retains a pivotal role, orchestrating the scientific inquiry while leveraging the AI to accelerate learning, refine mechanistic hypotheses, and execute complex bioinformatic analyses. By functioning as an intellectual partner rather than a mere tool, PersonaAI acts as a force multiplier. It enables a single scientist to execute multi-disciplinary workflows that traditionally required a large collaborative team, effectively compressing the timeline of discovery from months to days.

PersonaAI even has a capacity to suggest new hypothesis. A central question is if the RAG-based approach can produce truly novel hypotheses or if they are limited to summarizing and recombining existing knowledge. While critics might dismiss RAG as merely a sophisticated search mechanism (Bender et al., 2021; LeCun & Courant, 2022), advanced RAG architectures enable a powerful functional form of reasoning known as multi-hop inference (Trivedi et al., 2023). Besides, scientific novelty rarely arises from a complete scratch. It often emerges from connecting previously disparate concepts across domain boundaries. Our deployment of PersonaAI suggests that RAG-based architectures, when appropriately engineered, can indeed transcend mere information retrieval to drive genuine scientific novelty. The benchmarking we performed using temporal cut-off and the examples (Figure 2), clearly demonstrate that PersonaAI can suggest new hypothesis.

PersonaAI derives its novelty not by inventing biological principles *de novo*, but by functioning as a hyper-efficient engine for cross-disciplinary synthesis. Leveraging a RAG architecture that indexes over 560,000 aging-related publications, the system ingests a knowledge corpus that vastly exceeds human cognitive limits, spanning siloed disciplines from evolutionary biology to clinical immunology and neuroscience. In a key demonstration of this capability, PersonaAI conceptually linked the phenotypic signature of elevated CIRBP in human skeletal muscle to the induction of hepatic (liver) inflammation (Figure 3), which has not been previously investigated. Thus, the system's novelty emerges from its computational ability to detect latent semantic connections between disparate biological contexts that a single human specialist would be unlikely to bridge.

The capacity for novelty was demonstrated in our case study on ASPCs (Figure 4). PersonaAI did not simply report that ASPCs decline or that vasculature deteriorates. Instead, it synthesized these observations into a granular mechanistic model, proposing the specific uncoupling of the VEGF-VEGFR signaling axis as the primary driver of niche dysfunction. This moves from high-level association to a testable molecular mechanism represents a distinct value-add that goes beyond summarizing existing text.

Meta-prompting serves as the foundational governance layer of the PersonaAI architecture. This engineered prompting framework is critical for mitigating the stochasticity inherent in LLMs, ensuring that all generated outputs remain constrained within the bounds of biological plausibility. To achieve this, we developed a library of task-specific meta-prompts designed to impose rigid structural logic on the agent's reasoning. An example is our three-phase refinement protocol, which utilizes a "Step-Back" mechanism. This architecture compels the agent to first abstract the user's query into fundamental biological principles before progressively converging on precise, testable molecular targets (Figure 4).

Assessment using multi-agent is a crucial part in our system. By integrating agentic-AI for scRNA-seq analysis, PersonaAI can perform validation with enhanced flexibility. Rather than following a rigid script, the agent autonomously develops a customized strategy to test each biological hypothesis against the specific characteristics of the available data. Besides, it has a potential to be designed for multi-omics data. While our current focus is the biology of aging, the same agentic discovery loop could be deployed to uncover mechanisms across a wide spectrum of complex pathologies.

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Appendix A. Detailed Method

Data collection and preprocessing

1) Aging Atlas Repository

Our multi-agent system operates on comprehensive scRNA-seq datasets from the Mouse Aging Atlas (Sziraki et al., 2023; Zhang et al., 2025). The primary dataset comprises over 25 million cells from 17 tissue types (adipose: BAT, gWAT, iWAT; major organs: Brain, Heart, Kidney, Liver, Lung, Muscle; gastrointestinal: Stomach, Duodenum, Jejunum, Ileum, Colon; immune: B cells, T cells, Myeloid cells) spanning 5 age points from 3 to 23 months. Biological variables include sex (male/female) and strain-matched cohorts.

2) Data Storage and Preprocessing

All scRNA-seq data is stored using TileDB-SOMA (Scalable Open Matrix API) (Hoffman & Wolen, 2025), a cloud-optimized columnar storage format with three components: obs (cell-level metadata: cell type, age group, sex, QC metrics), var (gene metadata: ENSEMBL IDs, gene symbols), and X (sparse expression matrix). Data is organized into experiment-level partitions by tissue type with pre-annotated cell type labels from the original atlas construction, enabling SQL-like predicate pushdown filtering and parallel I/O operations without loading entire experiments into memory. Cell filtering supports complex Boolean logic queries on cell type, age group, sex, and QC thresholds, while gene filtering operates on gene symbols rather than ENSEMBL IDs. Three sampling strategies are available: random (uniform sampling), stratified (age-balanced), and first-N (deterministic). Extracted data undergoes automated preprocessing where ENSEMBL IDs are converted to gene symbols with duplicate resolution, genes are lexicographically sorted, and output is standardized to AnnData format (h5ad) compatible with Scanpy (Wolf et al., 2018).

Aging Hypothesis generation agent

To overcome the limitations of standard information retrieval, we designed the hypothesis generation module as a Self-Correcting RAG system driven by a LangGraph-based state machine ((Yan et al., 2024), (Ai, 2024), (Wang & Duan, 2024)). Unlike linear QA models, this system emulates the recursive reasoning process of a human researcher through an iterative "Generation-Critique-Refinement" loop.

1) Knowledge Base and Retrieval

The system indexes over 560,000 aging-related publications from PubMed/PMC. To ensure high-precision retrieval, we implemented a hybrid chunking strategy and a two-stage retrieval process involving query expansion and re-ranking (Flashrank). This infrastructure supports the "Knowledge Synthesis" phase described in Figure 1A.

2) Cognitive Prompt Framework (Epistemic Persona Injection)

To reduce the risk of unsupported generation, we impose explicit knowledge boundaries: hypothesis generation is restricted to pre-2021 literature, and the resulting claims are subsequently evaluated against held-out post-2021 studies and atlas-based evidence. This design is intended to mitigate hallucination risk, but it does not eliminate it; therefore, outputs are treated as hypotheses pending verification.

3) Hierarchical Hypothesis Construction

The generation process adapts to the complexity of the research question using three distinct strategies, fully detailed in Supplementary Table 1-7.

1. Temporal Cutoff Strategy

For retrospective validation, the agent was constrained to pre-2020 literature to predict post-2021

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findings (Supplementary Table 1 and 2).

2. Cross-Resolution Gap Finding

For the CIRBP case study, the agent used a "Bulk-to-Single-Cell" logic to identify cellular mechanisms underlying bulk tissue phenomena (Supplementary Table 3).

3. Meta-Prompting Architecture

The hypothesis generation process follows a structured, multi-step reasoning chain—comprising Trend Scouting, Idea Generation, and Hypothesis Finalization to construct granular mechanistic models.

Research Trend Sprouting: The agent acts on a broad research field to analyze the landscape and identify key unexplored themes, filtering out well-saturated topics to select one high-potential area.

Idea Generation: Focusing on the selected theme, the agent engages in constrained brainstorming to propose novel mechanistic narratives that explain the observed phenomena.

Hypothesis Finalization: Finally, the agent refines the abstract idea into a structured, scientifically testable prediction, explicitly defining the Source, Signal, Target, and Mechanism.

For complex queries like sex-dimorphism, this workflow is further intensified to ensure the multi-dimensional resolution of the proposed models (Supplementary Table 4-7).

Finally, generated hypotheses undergo a self-evaluation step where the system quantitatively assesses logical completeness (0.0-1.0 scale) using an LLM-based scoring rubric (Table 1). This rubric evaluates criteria such as coverage depth, integration quality, and theoretical basis. Only hypotheses surpassing a strict threshold (0.9) are passed to the Hypothesis Evaluation Agent for in-silico validation.

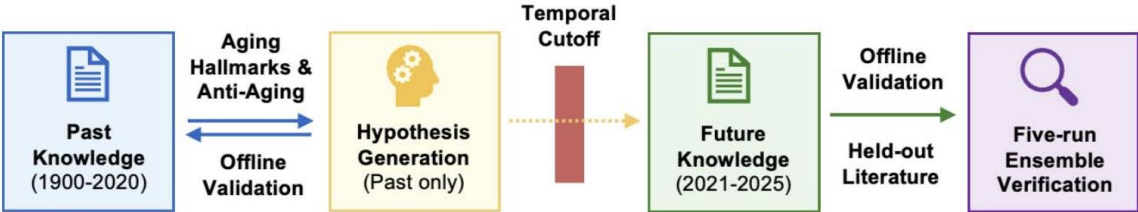
Statistical analysis:

For single-cell transcriptomic analysis, age-associated gene expression trends and differential expression were evaluated using the Wilcoxon rank-sum test. All computational procedures were executed through the PersonaAI framework, primarily utilizing Scanpy for data processing and statistical evaluation.

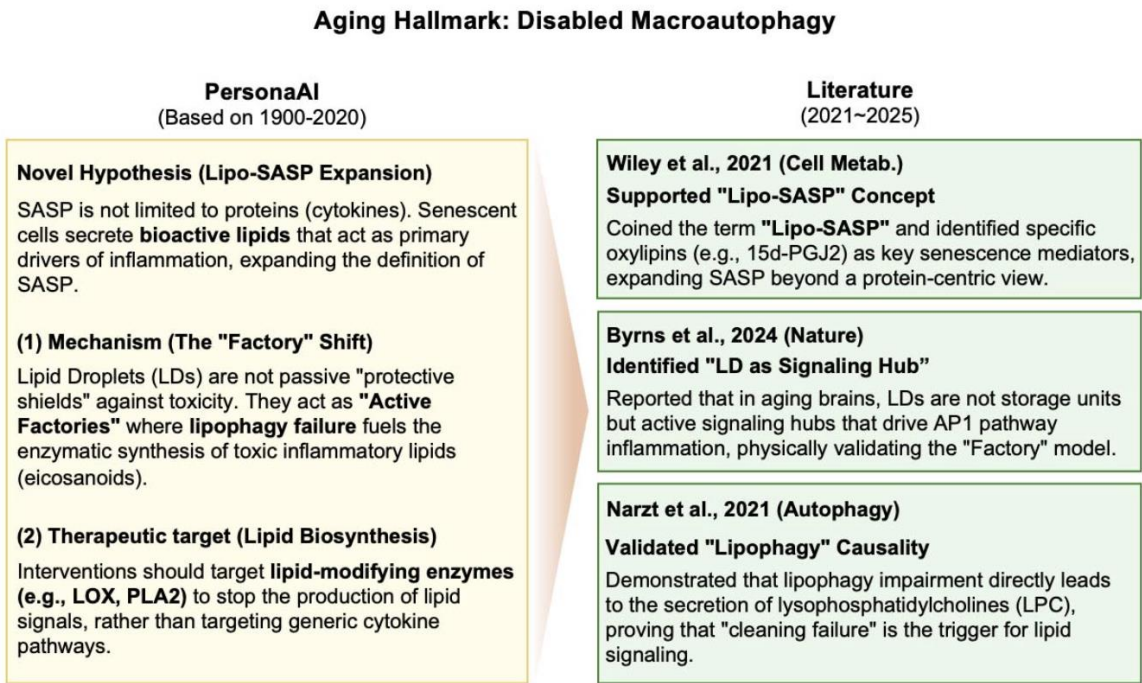
Data Availability

The biomedical literature data used for constructing the Knowledge Graph in this study were retrieved from the publicly accessible PubMed database (<https://pubmed.ncbi.nlm.nih.gov/>). The high-dimensional Knowledge Graph constructed from these abstracts and the associated source code are not publicly available at this stage to maintain anonymity during the double-blind review process. They will be made available in a public repository upon acceptance.

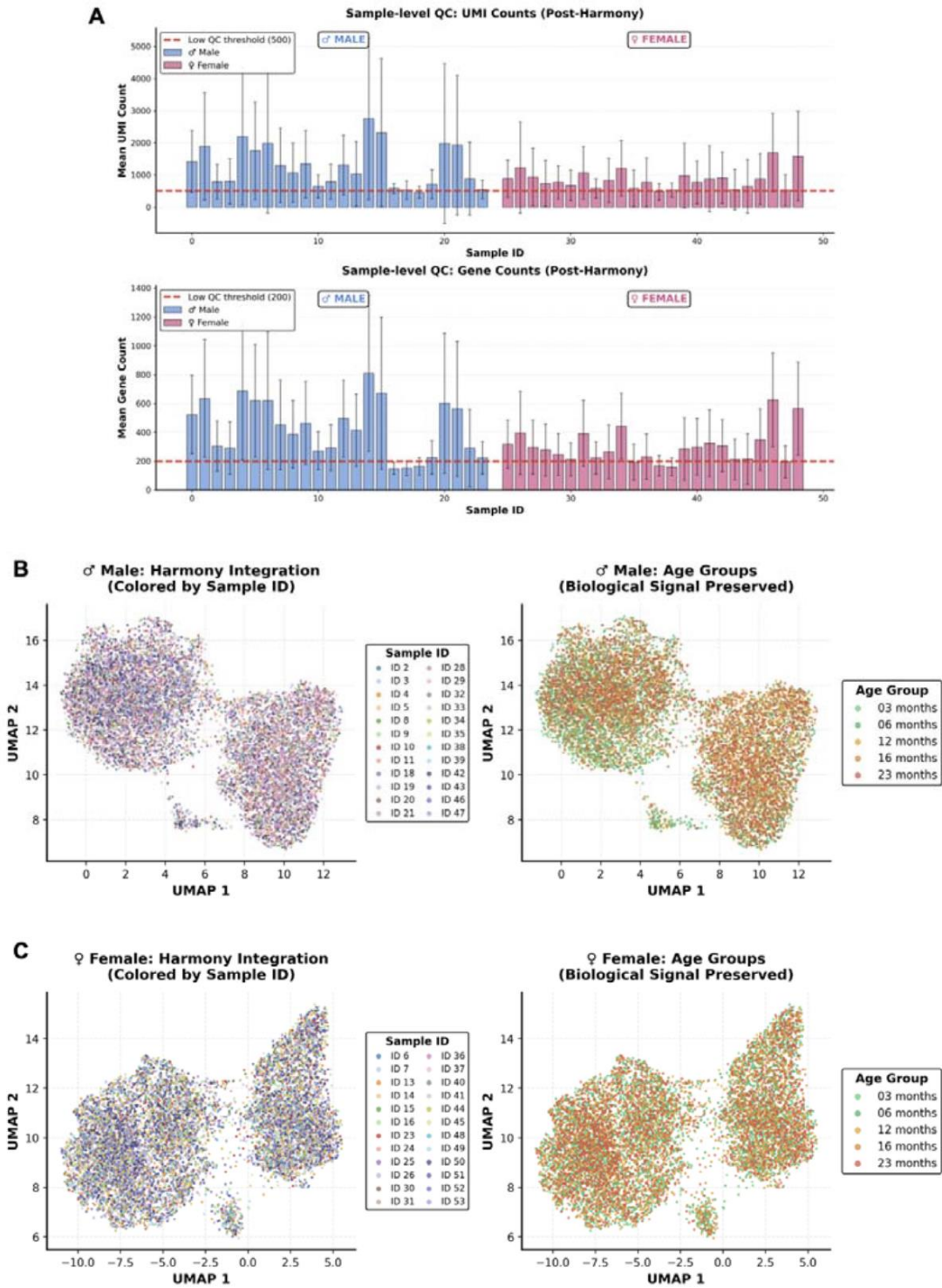
Appendix B. supplemental material



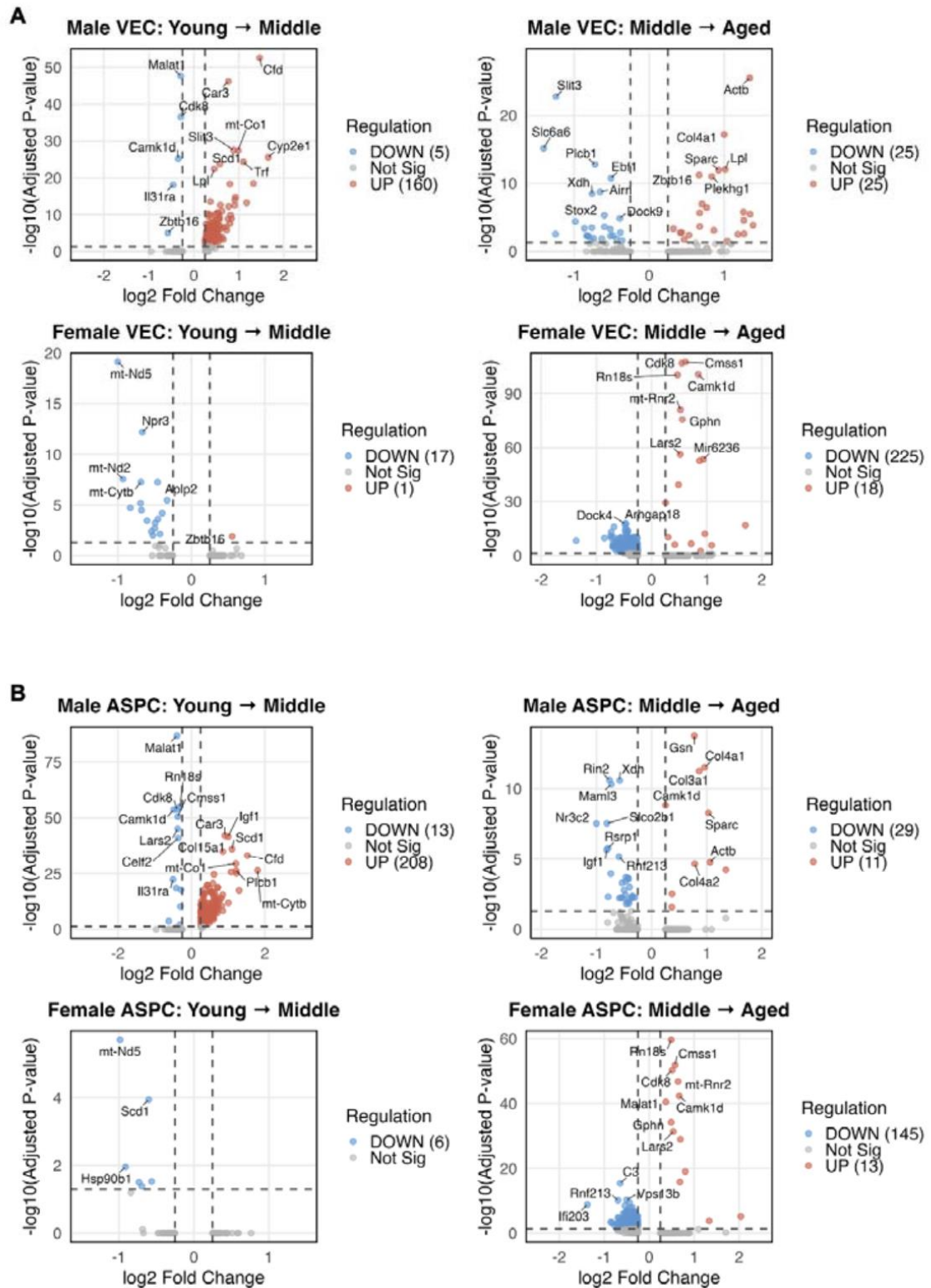
Supplementary Figure 1. Workflow of the Temporal Cutoff Validation Strategy.



Supplementary Figure 2. Generation and Validation of the Lipo-SASP Hypothesis.



Supplementary Figure 3. Single-cell Data Quality Control and Integration.



Supplementary Figure 4. Transcriptomic Aging Signatures in VECs and ASPCs.

Table 1. Evaluation Criteria for Assessing Hypothesis Relevance and Completeness

Criterion	Rating	Qualitative Descriptor
Relevance	0.0-0.3	Answer largely misses most main topics or addresses them superficially
	0.4-0.6	Answer addresses some main topics but has significant gaps in coverage
	0.7-0.8	Answer addresses most main topics well with minor gaps
	0.9-1.0	Answer comprehensively addresses ALL main topics with precision (RESERVE FOR TRULY EXCEPTIONAL COVERAGE)
Completeness	0.0-0.2	Superficial treatment of main topics, lacking depth
	0.3-0.5	Basic analysis of main topics with limited theoretical grounding
	0.6-0.7	Good analysis of main topics with some theoretical foundations
	0.8-0.9	Excellent analysis with strong theoretical grounding of main topics
	0.9-1.0	Exceptional analysis with profound insights into all main topics (RESERVE FOR WORLD-CLASS ANALYSIS ONLY)

Supplementary Table 1. Logic Specification for Prompt concept 1 (CIRBP)

Component	Implementation Details & Prompt Constraints
Epistemic Persona Injection	Single-Cell Validation Expert Persona Assigns the persona of an Aging Biology Expert who validates bulk research findings at the single-cell level. The goal is to make new discoveries by identifying phenomena reported in bulk literature but not yet verified at the single-cell resolution.
Fusion Logic (Core Task)	Cross-Resolution Gap Finding Protocol Bulk Signal Detection: Identify specific genes/proteins reported in learned literature to show significant age-related changes (increase/decrease) based on Bulk evidence. Verify Blindspot (Research Gap): Select only cases where the specific cell type driving these changes is unknown or not described in single-cell studies within the literature. Organ Mapping: Strictly limit the scope to one of the 14 Aging Atlas organs: BAT, Brain, Colon, Duodenum, Heart, Ileum, Jejunum, Kidney, Liver, Lung, Muscle, Stomach, gWAT, iWAT.
Constraints (Anti-Hallucination)	Exclusion & Priority Filters Novelty Priority (Avoid Famous Markers): Strictly exclude "hyper-famous markers" where cell types are already well-established. • <i>Cell Senescence</i>: CDKN2A (p16), CDKN1A (p21), TP53 • <i>Inflammation</i>: IL6, TNF, CCL2, CCL5, CXCL10 • <i>Microglia</i>: C1QA, C1QB, C1QC, TREM2, AIF1 • <i>Astrocytes</i>: GFAP, S100B, AQP4 • <i>Fibrosis</i>: COL1A1, COL3A1, ACTA2 Distribution Constraint: Must include at least 2 tissue-intrinsic phenomena; maximum 1 plasma-based phenomenon allowed. Focus: Focus on 1-3 genes per theme.
Output Protocol	Gap Verification Format Report 3 themes including the following 4 categories: - Theme: [Organ] - [Gene Name] (e.g., Heart - Timp1) - Bulk Evidence: Cite literature [Author Year] and specify the finding (Gene Increase/Decrease). - Unresolved Question: Formulate a question: "Which cell type/subtype drives the change in [Gene Name]?" - Aging Atlas Analysis Plan: Propose a method to compare expression by cell type and track age-related changes within the Aging Atlas.

Supplementary Table 2. Prompt concept 2 (Meta-Prompting/Sex-specific)

Step	Epistemic Persona Injection	Logic Specification & Prompt Constraints
Step 1: Research Trend Scouting	Sex-specific Aging Expert (Focus: Tissue Heterogeneity)	Asymmetry Detection Protocol Refuses simplistic hormone-centric explanations and identifies sexual asymmetry at the "Driver Cell" level. • Targeting: Sets a target tissue with distinct sex-specific aging patterns (e.g., Perigonadal Adipose Tissue). • Shift Focus: Endocrine Level → Cellular Dynamics Level. • Output: Derives 3 verifiable asymmetric research themes for the Aging Atlas.
Step 2: Idea Generation	Stem Cell Biologist (Focus: Mechanism Divergence)	Divergent Fate Mapping Expands the asymmetry identified in Step 1 into 5 categories of mechanistic ideas. • Anchor Logic: Generates hypotheses centering on the "Male-Specific Loss (of Protective Signal)" vs. "Female Maintenance" pattern. • 5-Category Constraints: Forces exploration across Communication, Pathways, Patterns, Heterogeneity, and Rejuvenation to broaden the scope. • Feasibility: Mandates the inclusion of literature hints (Indirect Evidence) and data verifiability.
Step 3: Hypothesis Finalization	Network Architect (Focus: Causal Integration)	Causal Triangulation Refines abstract ideas into verifiable Molecular Circuits to assemble the final hypothesis. • Triangulation: Combines three elements: [Who: Niche] - [What: Ligand-Receptor] - [Why: Loss Mechanism].

Supplementary Table 3. Logic Specification applied to Prompt concept 2 (Step 1)

Component	Implementation Details & Prompt Constraints
Epistemic Persona Injection	Sex-specific Aging & Adipose Biologist The system adopts the persona of a specialist in Sex-specific Aging and Visceral Adipose Biology. It possesses a deep understanding of cellular dynamics and OSKM reprogramming, transcending the traditional endocrinological perspective to analyze tissue-level aging heterogeneity.
Scouting Logic (Core Task)	Asymmetry Detection Protocol 1. Target Domain Definition: Focuses exclusively on "Perigonadal Adipose Tissue," a depot critical for metabolic disease that exhibits distinct aging trajectories between sexes (e.g., visceral accumulation patterns). 2. Shift Focus (Beyond Hormones): Refuses to attribute aging differences solely to hormonal variance. Instead, it aims to identify specific "Driver Cell Populations" that actively orchestrate the aging process within the tissue context. 3. Trend Analysis: Analyzes literature from the past 5 years to identify three high-impact biological themes and associated Research Gaps that have been overlooked.
Constraints (Context Awareness)	Single-Cell Validation Readiness Proposed themes must not remain abstract theories but must be verifiable using Aging Atlas single-cell data (scRNA-seq). The output must highlight opportunities to elucidate "Cell-Cell Communication (Cell-Chat)" networks or distinct "Cellular Subtypes" that drive sexual dimorphism.
OutputProtocol	Structured Theme Report (3 Themes) Each theme must be reported with the following four mandatory components: Core Concept: A summary of how aging manifests asymmetrically between sexes. Driver Cell Characteristics: Identification of the specific cell types driving this asymmetry and the rationale. Limitations of Current Research: A critique of existing studies, particularly the lack of rigorous sex-comparative analyses. Opportunities for Single-Cell Analysis: Specific points for validation, such as unique signaling pathways or rare cell populations.

Supplementary Table 4. Logic Specification applied to Prompt concept 2 (Step 2)

Component	Implementation Details & Prompt Constraints
Epistemic Persona Injection	Stem Cell & Single-Cell Biologist The system adopts the persona of an authority in single-cell transcriptomics and stem cell biology. It is tasked with going beyond descriptive observation to identify "Unexplored Territories" in sex-specific aging mechanisms, specifically focusing on the microenvironmental cues regulating stem cell exhaustion.
Generation Logic (Core Task)	Divergent Fate Mapping & Mechanism Transplantation 1. Input Context (Anchor): The logic is anchored on the key finding from Step 1: <i>"Early depletion and hypertrophy of male ASPCs versus the proliferative maintenance of female ASPCs."</i> 2. 5-Category Ideation Strategy: The system is constrained to generate hypotheses across five mandatory categories to ensure diverse mechanistic exploration: • Cat A (Communication): Novel cell-cell interaction pairs (e.g., Macrophage → ASPC signaling). • Cat B (Pathways): Overlooked non-canonical pathways (vs. canonical Wnt/BMP). • Cat C (Pattern): Communication reconfiguration patterns, specifically focusing on the "Loss of protective signals" in males. • Cat D (Heterogeneity): Sex-dimorphic composition of ASPC subtypes. • Cat E (Rejuvenation): Differential responsiveness to OSKM reprogramming interventions.
Constraints (Context Awareness)	Feasibility Check (Aging Atlas) Proposed ideas must be grounded in reality; the system must evaluate the "Verifiability" of each idea within the Aging Atlas dataset (e.g., detectability of specific ligand-receptor pairs). It is mandatory to provide "Literature Hints" (indirect evidence) to support biological plausibility.
Output Protocol	Structured Idea Proposal (5 Candidates) Each of the five generated ideas must adhere to the following format: 1. Research Question: Explicitly defines <i>Who</i> (Sex), <i>What</i> (ASPC/Signal), and <i>How</i> (Mechanism). 2. Why it matters: Articulates the impact and novelty compared to existing literature. 3. Literature Hints: Cites indirect evidence supporting the hypothesis. 4. Aging Atlas Feasibility: Assesses validation potential (Yes/Partial) with specific markers.

Supplementary Table 5. Logic Specification applied to Prompt concept 2 (Step 3)

Component	Implementation Details & Prompt Constraints
Epistemic Persona Injection	Network Architect & Verification Expert The system adopts the persona of an expert in single-cell data analysis and cell-cell communication networks. It is responsible for translating abstract ideas into concrete, verifiable molecular hypotheses.
Finalization Logic (Core Task)	Causal Triangulation & Integration Protocol 1. Input Context: Based on the "Male-Specific Loss of Female Protective Signals" paradigm selected in Step 2. 2. Triangulation (3-Axis Exploration): The system independently identifies three core components: • Who (Niche): Top 3 candidate cell populations (e.g., immune or vascular cells) that potentially send protective signals to ASPCs. • What (Signal): Specific Ligand-Receptor pairs mediating the support function (Proliferation/Anti-aging). • Why (Mechanism): The biological reason for signal termination in males (e.g., Inflammatory inhibition, Niche senescence). 3. Integration Strategy: The system synthesizes the optimal combination into a unified hypothesis using the following formula: $[Niche_X] \xrightarrow{Signal_Y} [ASPC_{Function}]$ where $\begin{cases} \text{Male: Loss via } Mechanism_M \\ \text{Female: Maintained (Protective)} \end{cases}$
Constraints (Verification)	Concrete Verifiability (Aging Atlas) The hypothesis must be actionable. It requires explicit validation points using Aging Atlas data, specifically predicting differential expression patterns of identified markers.
Output Protocol	Integrated Hypothesis Report The output must follow a strict format: 1. Candidate Pool: A list of candidates for Niche cells, Signaling pathways, and Loss mechanisms. 2. Final Hypothesis Statement: A declarative statement explaining that "Signal Y from Niche X supports ASPC Function Z, but is lost in males due to Mechanism M". 3. Verification Points: Three specific predictions verifiable via the Aging Atlas (e.g., "Check if Receptor R is downregulated in Male ASPCs").

AI Co-Scientist Challenge Korea Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: Yes

Justification:

All major claims presented in the abstract and introduction are systematically demonstrated in the Results section. Specifically: (1) the construction of PersonaAI's dual-phase agentic framework (Figure 1), (2) prospective validation of 114 hypotheses through temporal cutoff strategy (Figure 2), (3) discovery of senescent CIRBP+ hepatocytes (Figure 3), and (4) identification of vascular niche mechanisms driving male-specific ASPC decline (Figure 4) are clearly substantiated with comprehensive experimental evidence.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: Yes

Justification:

The Discussion section provides a balanced discourse on the inherent characteristics and boundaries of RAG-based systems. We explicitly describe design features implemented to mitigate hallucination risks, including epistemic persona injection and a stringent self-evaluation threshold (0.9). Furthermore, we transparently define the scope of our system by characterizing outputs as "hypotheses pending verification," ensuring appropriate expectations for downstream use.

3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: NA

Justification:

This paper is an applied research study focused on the design and empirical validation of an agentic AI framework for biological hypothesis generation and validation, rather than presenting mathematical theorems or formal proofs.

4. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: NA

Justification:

Same as 3 above.

5. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: Yes

Justification:

We provide comprehensive methodology to ensure reproducibility. Appendix A details: (1) data storage and preprocessing pipelines using TileDB-SOMA, (2) Self-Correcting RAG system architecture with hybrid chunking and two-stage retrieval, and (3) the three-phase hierarchical hypothesis generation protocol. Supplementary Tables 1-5 fully disclose all prompt logic specifications and constraints. All public data sources (Mouse Aging Atlas, PubMed/PMC) are explicitly cited with proper references.

6. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: NA

Justification:

Same as 3 above.

7. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: Yes

Justification:

Same as 5 above.

8. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: Yes

Justification:

This study prioritizes reproducibility and transparency. (1) Core single-cell data utilizes the publicly available Mouse Aging Atlas (Zhang et al., 2025; Sziraki et al., 2023), (2) literature data was collected from openly accessible PubMed/PMC databases. The Knowledge Graph and source code are currently withheld solely to maintain anonymity during the double-blind review process; we explicitly commit to releasing all assets in a public repository immediately upon acceptance.

9. Experimental Setting/Details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: Yes

Justification:

We provide detailed documentation for complete experimental reproduction. (1) Data structure: TileDB-SOMA format with obs/var/X components, (2) RAG system: hybrid chunking strategy with two-stage retrieval (query expansion + Flashrank re-ranking), (3) Evaluation criteria: quantitative rubric in Table 1 (Relevance/Completeness on 0.0-1.0 scale), (4) Statistical analysis: Wilcoxon rank-sum test specification. Supplementary Tables 1-5 fully disclose prompt components and constraints for each experiment.

10. Experiment Statistical Significance

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Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: Yes

Justification:

We employed a multi-layered validation strategy to ensure statistical rigor. (1) Hypothesis verification: ensemble validation across 5 independent runs with High/Medium/Unverified confidence classification (Figure 2A), (2) Single-cell analysis: significance testing via Wilcoxon rank-sum test (Figure 3B: $p < 0.00001$; Figure 4D: $p = 0.039$), (3) Quality control: batch correction validation and biological signal preservation verification demonstrated in Supplementary Figures 3-4.

11. Experiments Compute Resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: Yes

Justification:

PersonaAI was designed with accessibility and scalability as core principles. The framework leverages standard cloud-based LLM APIs and publicly available single-cell analysis tools (Scanpy), enabling reproduction in general research environments without requiring specialized high-performance computing infrastructure. This design philosophy aligns with our vision of democratizing AI co-scientists for the broader research community.

12. Code Of Ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://nips.cc/public/EthicsGuidelines>?

Answer: Yes

Justification:

This research adheres to all principles of ethical research conduct. (1) Exclusive use of publicly available biomedical literature (PubMed/PMC) and previously approved animal study data (Mouse Aging Atlas), (2) No use of sensitive personal information or human subject data, (3) Research objectives focused on socially beneficial directions: understanding aging mechanisms and extending healthy lifespan, (4) Proper citation and attribution of all data sources and tools utilized.

13. Broader Impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: Yes

Justification:

The Discussion section addresses the societal implications of PersonaAI from multiple perspectives. **Positive impacts:** (1) Serves as a "force multiplier" enabling individual researchers to conduct multi-disciplinary workflows comparable to large collaborative teams, (2) Compresses discovery timelines from months to days, (3) Provides access to complex bioinformatic analyses without specialized coding expertise. **Responsible use:** We explicitly characterize system outputs as "hypotheses pending verification" to prevent over-interpretation of AI results and maintain the importance of expert judgment through our human-in-the-loop design philosophy.

14. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: Yes

Justification:

Multiple safeguards for responsible AI development are embedded in our system design. (1) **Epistemic Persona Injection**: Explicitly constrains knowledge boundaries to prevent unfounded claims, (2) **Self-evaluation stage**: Only hypotheses surpassing a stringent threshold (0.9) proceed to validation, (3) **Human-in-the-loop design**: Collaborative reasoning with researchers ensures biological plausibility rather than fully autonomous operation, (4) **Temporal cutoff validation**: Provides an objective methodology to verify the system's inferential capabilities versus mere information retrieval.

15. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: Yes

Justification:

Proper citation and attribution are provided for all utilized assets. (1) **Datasets**: Mouse Aging Atlas (Zhang et al., 2025; Sziraki et al., 2023), PubMed/PMC public literature, (2) **Analysis tools**: Scanpy (Wolf et al., 2018), LangGraph (Ai, 2024), TileDB-SOMA (Hoffman & Wolen, 2025), LIANA+ (Dimitrov et al., 2024), GSEAPy (Fang et al., 2022), (3) **Methodological foundations**: RAG architecture, Step-Back Prompting (Zheng et al., 2024). All references are comprehensively listed in the Reference section.

16. New Assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: Yes

Justification:

All components of the PersonaAI framework are systematically documented. (1) **System architecture**: Complete workflow visualization in Figure 1, (2) **Prompt design**: Full disclosure of logic specifications, constraints, and output protocols in Supplementary Tables 1-5, (3) **Evaluation criteria**: Hypothesis assessment rubric presented in Table 1, (4) **Data pipeline**: End-to-end description from preprocessing to analysis in Appendix A. Complete source code and Knowledge Graph will be made available in a public repository upon acceptance.

17. Crowdsourcing and Research with Human Subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: NA

Justification:

This study does not involve crowdsourcing or human subject research. All analyses were conducted using existing public literature data and animal model-based single-cell datasets.

18. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human

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Subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: NA

Justification:

This research does not involve human subjects and does not include any research activities requiring IRB approval. All data used consists of (1) publicly available biomedical literature (PubMed/PMC), and (2) publicly released datasets from mouse studies (Mouse Aging Atlas) that were conducted under appropriate ethical approvals in their original studies.